

Cybersecurity Incident Report

Network Traffic Analysis — Day 22

Part 1: Summary of the Problem

Field	Analysis
UDP protocol reveals that:	The UDP protocol was used by the browser to communicate with the DNS server to retrieve the IP address for yummyrecipesforme.com. The DNS request received an ICMP error indicating that UDP port 53 was unreachable.
The port noted in the error message is used for:	DNS services and DNS protocol communications.
The most likely issue is:	The DNS server is unreachable or not responding to DNS requests.

Part 2: Analysis and Likely Cause

Field	Analysis
Time incident occurred:	1:24 PM (first log sequence: 13:24:32.192571).
How the IT team became aware:	Customers reported that they could not access yummyrecipesforme.com and received the error message “destination port unreachable” after waiting for the page to load.
Actions taken to investigate:	The cybersecurity team conducted packet-sniffing tests using tcpdump while attempting to access the website and analyzed the resulting network traffic.
Key findings:	The captured traffic showed UDP packets being sent to the DNS server and ICMP error responses indicating that UDP port 53 was unreachable. Since port 53 is associated with DNS traffic, the DNS service was not responding normally.
Next troubleshooting steps:	Determine whether the DNS server is functioning properly. If it is, check the firewall configuration to determine whether traffic to port 53 is being blocked.
Likely cause:	The DNS server may have become unavailable due to a suspected Denial of Service (DoS) attack that overwhelmed it with traffic. A firewall misconfiguration should also be investigated as an alternative cause.

Analyst conclusion: The evidence confirms that DNS traffic to UDP port 53 was unreachable. The available evidence does not by itself prove that a DoS attack occurred; it supports investigating both DNS server availability and firewall configuration, with a DoS attack considered a possible cause.